



DSP Integrated Circuits

Chapter 5: DSP Architecture

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- ◇ DSP Architecture Tradeoff Issues
 - Speed of execution, accuracy
 - ◇ Characteristics of DSP Processor
 - ◇ Pipelining & Parallelism
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 - ◇ Commercial DSP Processors
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DSP Architecture Tradeoff Issues

DSP Architecture Tradeoff issues

- ◇ Speed of execution
 - SW vs. hardwired logic
 - On-Chip multiplier
 - Pipelining & parallel execution for higher throughput
- ◇ Accuracy
 - Bit precision: More bits for higher accuracy
 - Floating point vs. fixed point
 - Quantization error
- ◇ Cost
 - Chip size
 - Power consumption
 - Program size (Memory)



Computing-speed Performance

- ◇ Benchmark speed criteria (Figure of merit for speed)
 - MIPS: Millions of Instruction per second
 - Measures instruction execution rate
 - Not useful because of different kinds of instructions
 - MFLOPS: Million Floating-point Operations Per Second
 - Widespread figure of merit in comparing floating-point processor



Factors affecting Computation Speed

- ◇ Computation efficient algorithm
- ◇ Novel system architecture
 - General-purpose architecture vs. DSP architecture
- ◇ Processor speed
- ◇ Memory access time
- ◇ Address generation time
- ◇ Selection of fabrication process



Accuracy Issues

◇ Fixed-point and floating-point formats

■ Fixed-point: Format F(b.c) for unsigned numbers where

● b: total # of digits,

● c: # of digits which lies to the right of the decimal point

➔ limitation of dynamic range

● ex)

	Mantissa	Exponent	
Largest	1111 1111 1111	1111	$= (2^{12} - 1) \times 2^{15}$
Smallest	0000 0000 0001	0000	$= 1 \times 2^0$

■ Floating point: $y = m \cdot 10^E$

● where m: mantissa, E: exponent

➔ Need pre-scale to avoid overflow/underflow

Precision vs. Dynamic range

- ◇ In fixed-point numbers, word length determines both the precision and dynamic range
- ◇ In floating-point numbers, mantissa sets precision and the exponent determines dynamic range
- ◇ Dynamic range grows linearly in fixed-point, and exponentially in floating-point

Word length		16	32	64
Fixed point	Dynamic Range	6.6×10^4	4.3×10^9	1.8×10^9
	Maximum Precision	> 4 digits	> 9 digits	> 19 digits
Floating point	$y = m \cdot 10^E$	(4E,12M)	(8E,24M)	(11E,51M)
	Dynamic Range	3.3×10^4	5.8×10^{76}	10×10^{1419}
	Mantissa Precision	> 3 digits	> 7 digits	> 15 digits





Characteristics of DSP Processor

DSP algorithms shape DSPs (1)

- ◇ Very computationally demanding
- ◇ Requires attention to numeric fidelity
- ◇ High memory bandwidth requirements
- ◇ Streaming data-and lots of it
- ◇ Predictable data access patterns
- ◇ Execution-time locality
- ◇ Math-centric
- ◇ Real-time constraints
- ◇ Standards: algorithms, interfaces



DSP algorithms shape DSPs (2)

- Computational demands → Multiple parallel execution units, hardware acceleration of common DSP functions
- Numeric fidelity → Accumulator registers, guard bits, saturation hardware
- High memory bandwidth → Harvard architecture, support for parallel moves
- Predictable data access patterns → Specialized addressing modes, e.g., modulo, bit-reversed



DSP algorithms shape DSPs (3)

- Execution-time locality → Hardware looping, streamlined interrupt handling
- Math-centricity → Single-cycle multiplier(s) or MAC unit(s), MAC instruction
- Real-time constraints → Few dynamic features, on-chip SRAM instead of cache
- Standards → 16-bit data types; rounding, saturation modes



Common features of DSP processors (1)

- ◇ Data path configured for DSP
- ◇ MAC (Multiply and Accumulation) block
- ◇ Specialized instruction set
- ◇ Multiple memory banks and buses
- ◇ Specialized addressing modes
- ◇ Specialized execution control
- ◇ Specialized peripherals for DSP



Common features of DSP processors (2)

◆ Data Path

DSP Processor

- Specialized hardware performs all key arithmetic operations in 1 cycle.
- MAC operation in 1 cycle
- Hardware support for managing numeric fidelity:
 - Shifters
 - Guard bits
 - Saturation

General-Purpose Processor

- Multiplies often take >1 cycle
- Shifts often take >1 cycle
- Other operations (e.g., saturation, rounding) typically take multiple cycles



Common features of DSP processors (3)

◆ Instruction Set

DSP Processor

- Specialized, complex instructions
- Multiple operations per instruction

```
mac x0,y0,a  x:(r0)+,x0  y:(r4)+,y0
```

General-Purpose Processor

- General-purpose instructions
- Typically only one operation per instruction

```
mov *r0,x0  
mov *r1,y0  
mpy x0,y0,a  
add a,b  
mov y0,*r2  
inc r0  
inc r1
```

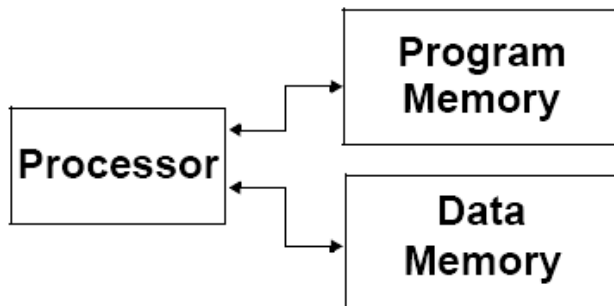


Common features of DSP processors (4)

◇ Memory Architecture

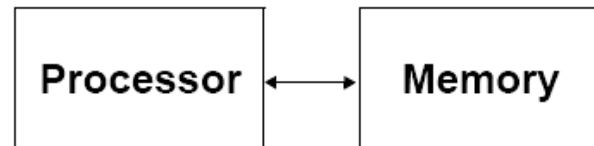
DSP Processor

- Harvard architecture
- 2-4 memory accesses/ cycle
- No caches—on-chip SRAM



General-Purpose Processor

- Von Neumann architecture
- Typically 1 access/ cycle
- May use caches



Common features of DSP processors (5)

◆ Addressing

DSP Processor

- Dedicated address generation units
- Specialized addressing modes; e.g.:
 - Autoincrement
 - Modulo (circular)
 - Bit-reversed (for FFT)
- Good immediate data support

General-Purpose Processor

- Often, no separate address generation unit
- General-purpose addressing modes



Common features of DSP processors (6)

◇ Execution Control

- Hardware support for fast looping
- “Fast interrupts” for I/O handling
- Real-time debugging support



Common features of DSP processors (7)

◆ Specialized Peripherals for DSPs

- Synchronous serial ports
 - Parallel ports
 - Timers
 - On-chip A/D, D/A converters
 - Host ports
 - Bit I/O ports
 - On-chip DMA controller
 - Clock generators
-
- On-chip peripherals often designed for “background” operation, even when core is powered down.



DSP vs Microprocessors (1)

- ◇ **Microprocessors: General-Purpose Processors (GPPs)**
 - CPUs for PCs and workstations (CISC-type)
 - E.g., Intel Pentium III
 - 32-bit GPPs for embedded applications (RISC-type)
 - E.g., ARM ARM7
- ◇ **Digital Signal Processors (DSPs)**
 - Microprocessors specialized for signal processing applications
- ◇ **Low-end DSPs and GPPs**
 - Architectures targeting extremely cost sensitive markets, often older architectures
- ◇ **High Performance DSPs and GPPs**
 - Architectures that use advanced techniques to improve parallelism, performance: Multi-issue architectures such as VLIW and superscalar
 - Usually have higher clock rates



DSP vs Microprocessors (2)

◇ CISC vs. RISC (1)

	CISC	RISC
Number of Instructions	A few hundreds	A few tens
Instruction Length	Variable (1-17 bytes)	Fixed (4 bytes)
Instruction Execution time	Variable (1-300 clock/inst.)	Fixed (1 clock/inst.)
Pipelining	Not efficient	Efficient



DSP vs Microprocessors (3)

◇ CISC vs. RISC (2)

	CISC	RISC
Control	Microcode	Hardwired
Number of Registers	Small (about 8)	Large (32-200)
Data memory Access	Memory Operands	Load, store Only
Design Effort	5-10x 600 (man/year)	1x 60 (man/year)



DSP vs Microprocessors (4)

◇ CISC vs. RISC (3)

■ Instructions

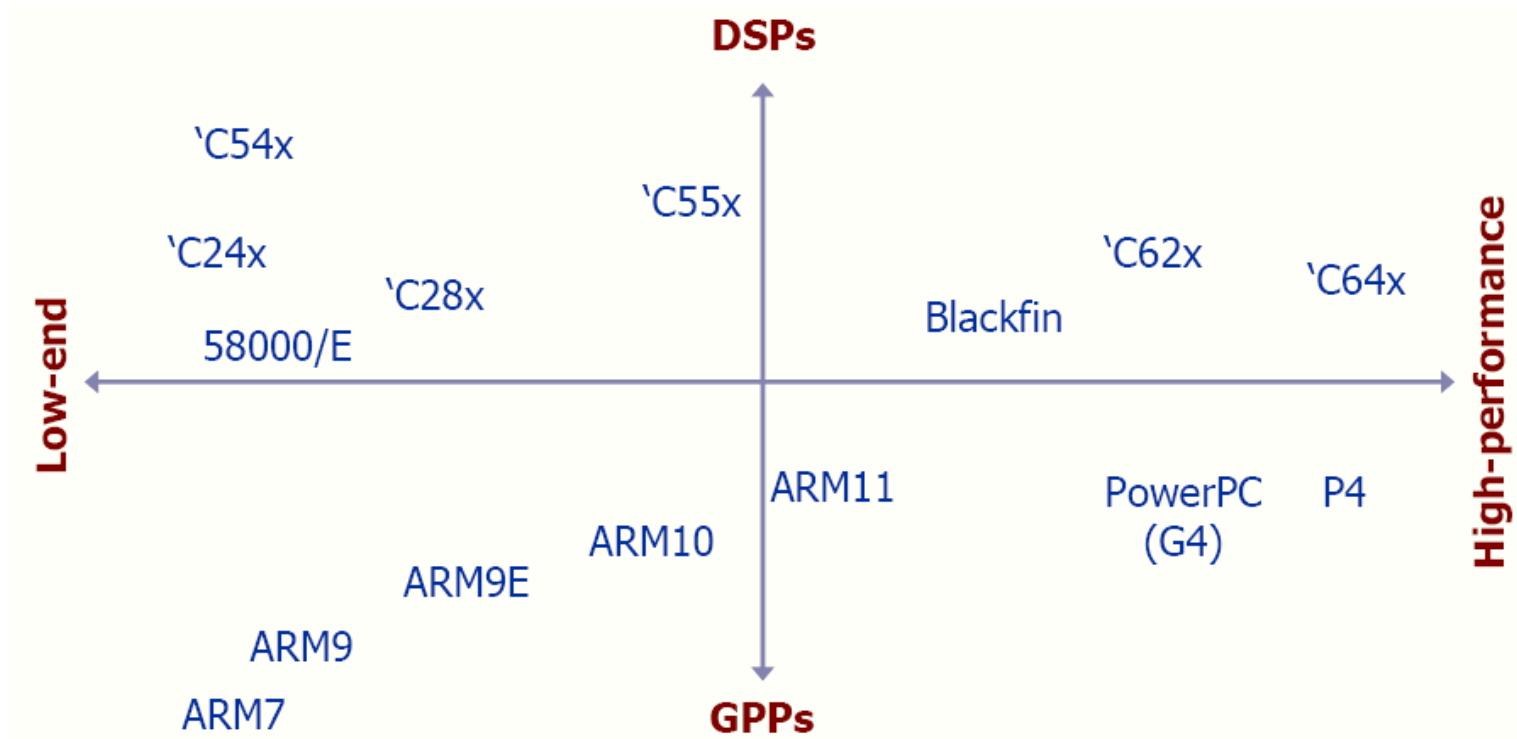
$$\begin{array}{l} \text{Mem} \leftarrow R_1 + \text{Mem} \\ = \left\{ \begin{array}{ll} \text{Temp}_1 \leftarrow \text{Mem} & R_2 \leftarrow \text{Mem} \\ \text{Temp}_2 \leftarrow R_1 + \text{Temp}_1 & R_3 \leftarrow R_1 + R_2 \\ \text{Mem} \leftarrow \text{Temp}_2 & \text{Mem} \leftarrow R_3 \end{array} \right. \end{array}$$

CISC microinstruction
= RISC instruction



DSP vs Microprocessors (5)

◇ Example Processors & DSPs

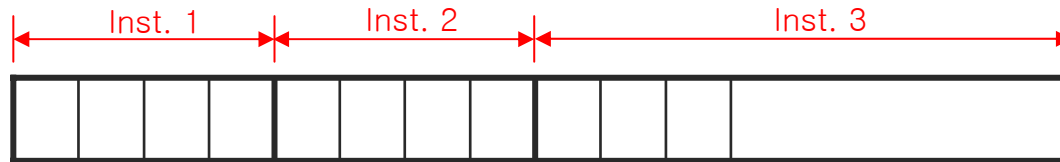




Pipelining & Parallelism

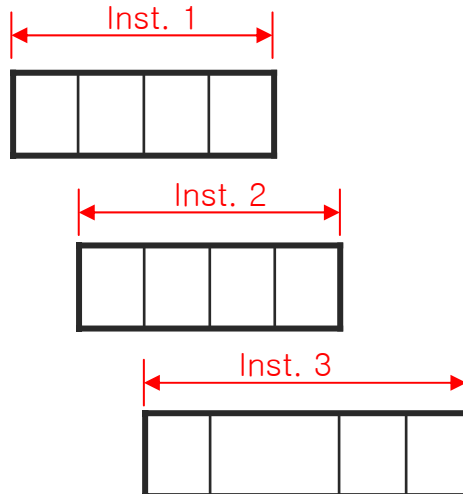
Pipelining & Parallelism(1)

◆ Conventional serial process:



◆ Pipelining process:

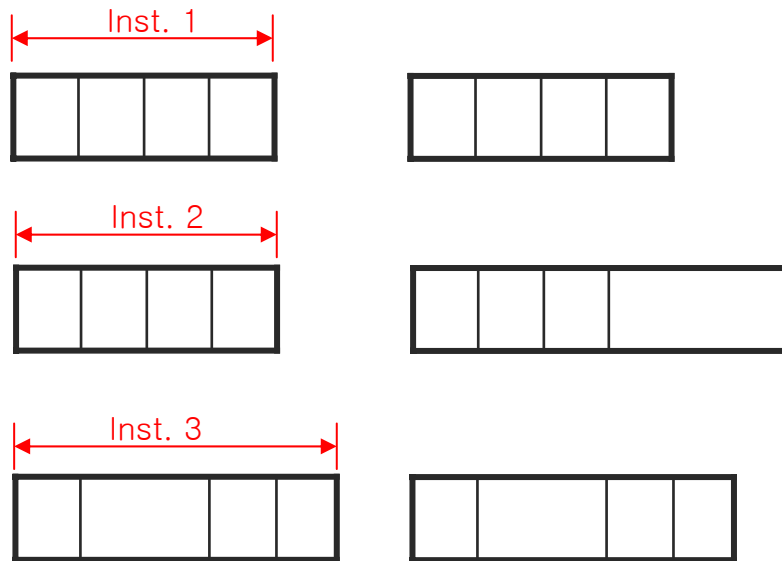
- Break each instruction into a sequence of separate operations, each of equal length



Pipelining & Parallelism(2)

◇ Parallel process

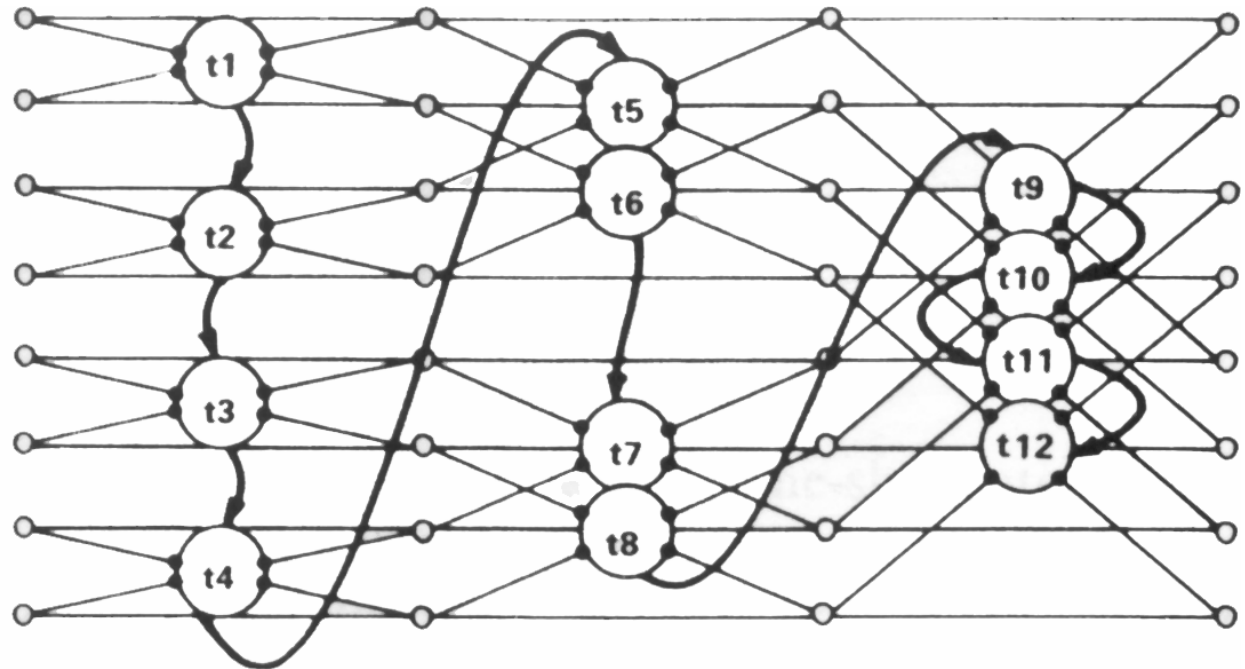
- Use several number of identical processing unit



Example of FFT Design(1)

Serial processing of FFT

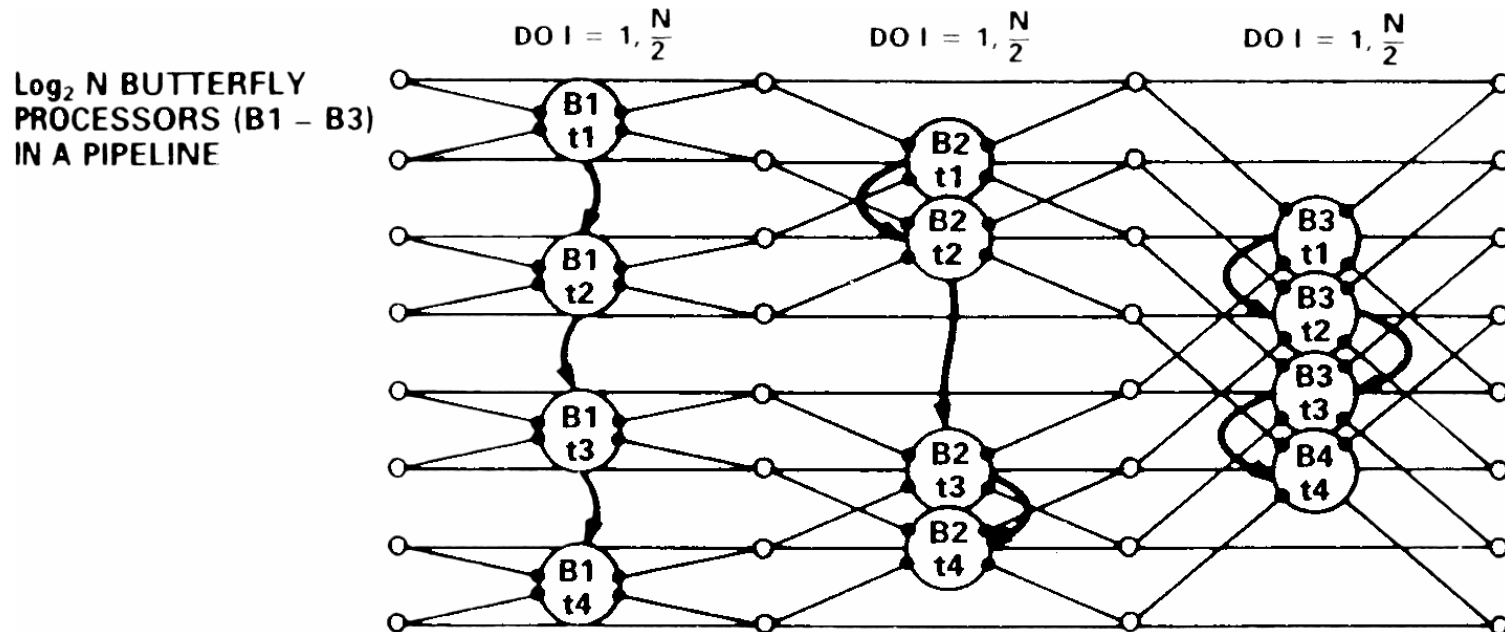
```
DO 20 J = 1, log2N  
DO 10 I = 1,  $\frac{N}{2}$   
10 CONTINUE  
20 CONTINUE
```



- Single BF processor
- Each BF is computed, one step at a time

Example of FFT Design(2)

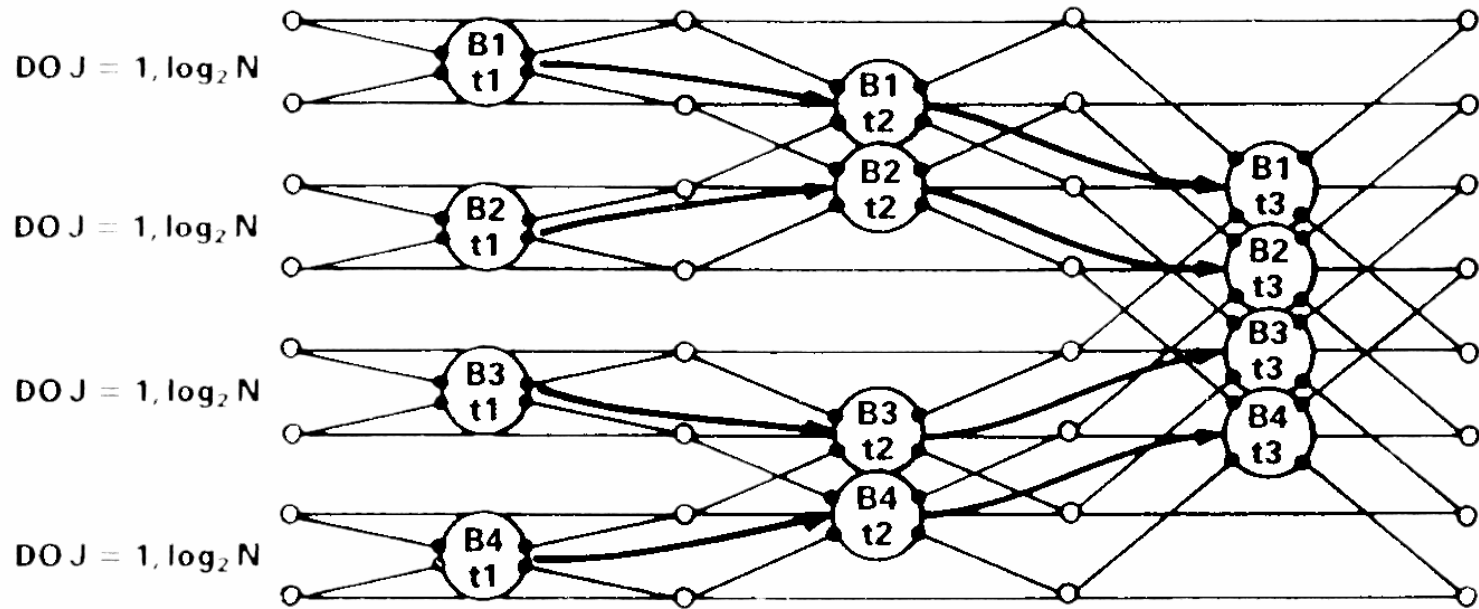
◆ Pipeline processing of FFT



- There are $\log_2 N$ BF processor (Corresponding to the number of passes)
- As each pass is computed, BFs are ready to accept a pair of inputs for the next pass

Example of FFT Design(3)

◇ Parallel processing of FFT $\frac{N}{2}$ BUTTERFLY PROCESSORS (B1 – B4) IN PARALLEL



- There are $N/2$ BF processors
- As soon as one pass is completed, all are ready for the next step

Summary of FFT performance

◇ FFT Speed Enhancement

Architecture	# of BFs	Computation time
Single processor	1	$N/2 \log_2 N$
Pipelined processors	$\log_2 N$	$N/2$
Parallel processors	$N/2$	$\log_2 N$

◇ ex) $N = 1024$

Architecture	# of BFs	Computation time
Single processor	1	5,120
Pipelined processors	10	512
Parallel processors	512	10



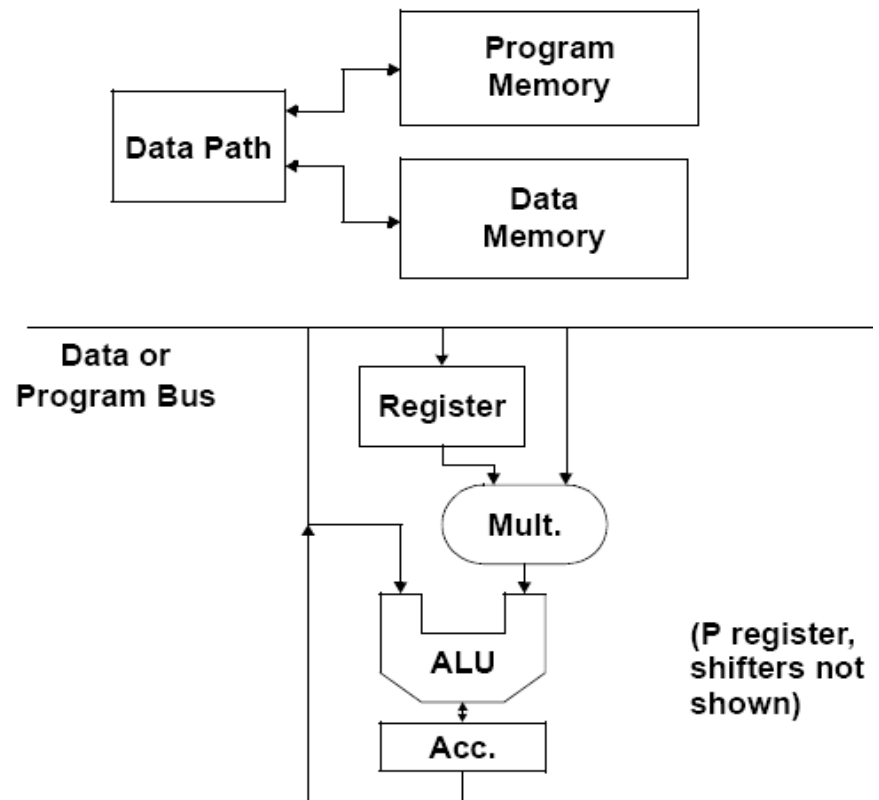
Evolution of DSP architecture

Evolution of DSP architecture (1)

◇ First Generation DSP (1982):

Texas Instruments TMS32010

- 16-bit fixed-point
- Harvard architecture
- Accumulator
- Specialized instruction set
- 390 ns MAC time (228 ns today)

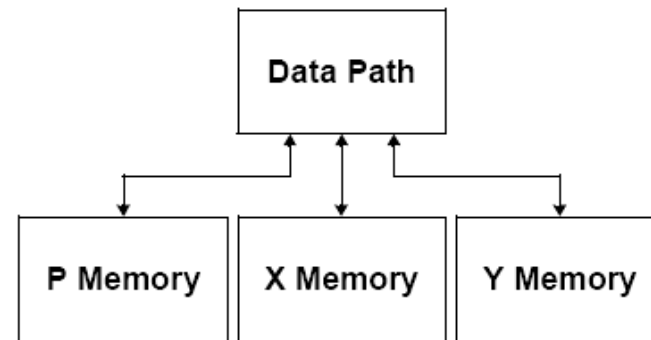


Evolution of DSP architecture (2)

◇ Second Generation DSPs (1987) Example:

Motorola DSP56001

- 24-bit data, instructions
- 3 memory spaces (X, Y, P)
- Parallel moves
- Single- and multi-instruction hardware loops
- Modulo addressing
- 75 ns MAC (21 ns today)



```
move  #Xaddr, r0
move  #Haddr, r4
rep    #Ntaps
mac    x0, y0, a    x: (r0)+, x0    y: (r4)+, y0
```

- Other second-generation processors: AT&T DSP16A, Analog Devices ADSP-2100, Texas Instruments TMS320C50



Evolution of DSP architecture (3)

◇ Third Generation DSPs (1995) Examples:

Motorola DSP56301, TI TMS320C541

- Enhanced conventional DSP architectures
- 3.0 or 3.3 volts
- More on-chip memory
- Application-specific function units in data path or as co-processors
- More sophisticated debugging and application development tools
- DSP cores (Pine and Oak from DSP Group, cDSP from TI)
- 20 ns MAC (10 ns today)

Architectural innovation mostly limited to adding application-specific function units and miscellaneous minor refinements.

- Also, multiple processors/chip (TI TMS320C80, Motorola MC68356)



Evolution of DSP architecture (4)

◇ Forth Generation DSPs Examples:

TI TMS320C6201, Intel Pentium with MMX

Today's top DSP performers adopt architectures far different from conventional DSP processor designs.

- Blazing clock speeds and superscalar architectures make some general-purpose processors, such as the PowerPC 604e, top floating-point performers, *despite lack of many DSP features*
- Multimedia SIMD extensions, such as MMX, offer strong fixed-point performance on general-purpose processors
- *But strong DSP tools for general-purpose processors are lacking*
- VLIW-like architectures, such as that of the TI TMS320C6201, achieve top performance via high parallelism and increased clock speeds (containing up to 8 separate mini instructions)
- 3 ns MAC throughput... but expensive, power-hungry



Evolution of DSP architecture (5)

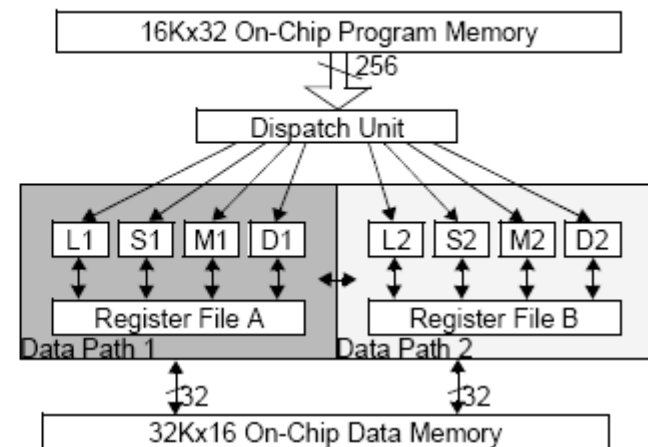
◇ VLIW (1)

Very long instruction word (VLIW) architectures are garnering increased attention for DSP applications.

Notable recent introductions include Texas Instruments' TMS320C62xx and Philips' TM1000.

Major features:

- Multiple independent operations per cycle
 - Packed into a single large “instruction” or “packet”
- More regular, orthogonal, RISC-like operations
- Large, uniform register sets



Evolution of DSP architecture (6)

◇ VLIW (2)

Advantages:

- Increased performance
- More regular architectures
 - Potentially easier to program; better compiler targets
- Scalable?

Disadvantages:

- New kinds of programming/compiler complexity
- Code size bloat
 - High program memory bandwidth requirements
- High power consumption

Evolution of DSP architecture (7)

◇ General-Purpose Processors are Catching Up

“Go where the cycles are...”

General-purpose processors are increasingly adding DSP capabilities via a variety of mechanisms:

- Add single-instruction, multiple-data instruction set extensions (e.g., MMX Pentium)
- Integrate a fixed-point DSP processor-like data path and related resources with an existing $\mu\text{C}/\mu\text{P}$ core (e.g. Hitachi SH-DSP)
- Add a DSP co-processor to an existing $\mu\text{C}/\mu\text{P}$ core (e.g., ARM Piccolo)
- Create an all-new, hybrid architecture (e.g., Siemens TriCore)



Evolution of DSP architecture (8)

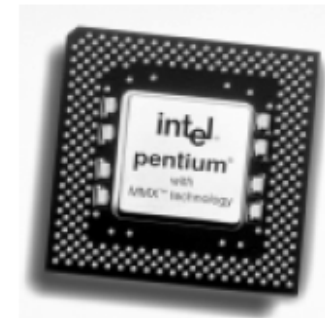
◆ The General-Purpose Processor Threat

High-performance general-purpose processors for PCs and workstations are increasingly suitable for some DSP applications.

- E.g., Intel MMX Pentium, Motorola/IBM PowerPC 604e

These processors achieve excellent to outstanding floating- and/or fixed-point DSP performance via:

- Very high clock rates (~ 3 GHz)
- Superscalar (“multi-issue”) architectures
- Single-cycle multiplication and arithmetic ops.
- Good memory bandwidth
- Branch prediction
- In some cases, single-instruction, multiple-data (SIMD) ops.



Evolution of DSP architecture (9)

◆ High-Performance General-Purpose Processors

Advantages:

- Strong DSP performance
- Already present in PCs
- Strong tool support for the major processors
- Cost-performance can rival that of floating-point DSPs



Disadvantages:

- Lack of execution timing predictability
- Difficulty of developing optimized DSP code
- Limited DSP-oriented tool support
- High power consumption
- Cost-performance does not approach that of fixed-point DSPs

Evolution of DSP architecture (10)

◇ Real-Time Suitability

The most important DSP applications are real-time applications.

- Many of these are “hard real-time” applications: failure to meet a real-time deadline creates a serious malfunction.

High-performance GPPs make heavy use of dynamic features:

- Caches, branch prediction, dynamic superscalar execution, data-dependent instruction execution times, etc.

These features result in timing behavior that appears to be stochastic.

- This seriously complicates development of DSP applications.

PC applications are further complicated by the lack of real-time support in PC operating systems.



The title is framed by two thick, curved lines. A light purple line starts on the left, curves upwards and then downwards, ending under the word 'Processors'. A blue line starts above the word 'Processors', curves upwards and then downwards, ending on the right.

Commercial DSP Processors

Commercial DSP Processors

Analog Devices	ADSP-21xxx (SHARC) family
	ADSP-BF53x (Blackfin) family
Motorola	DSP563xx family
	DSP5685x family
	MSC81xx family
Texas Instruments	TMS320C55x
	TMS320C62x
	TMS320C64x
	TMS320C67x



Embedded DSP Cores

- ◇ OakDSP
- ◇ TeakDSP
- ◇ PalmDSP
- ◇ JazzDSP
- ◇ ZSP500 (LSI Logic)
- ◇ ARC configurable core (600, 700)
- ◇ ARM piccolo



Embedded DSP Cores (1)

◇ OakDSP core (1)

- 16bit fixed-point DSP core based on single MAC architecture
- Designed for mid-to-high-end telecommunications and consumer electronics applications.
- Features
 - Performance (operating speed)
 - 80MIPS at 160MHz on 0.25um CMOS 2.5V
 - Low power consumption
 - 1mW/MIPS on 0.25um CMOS 2.5V
 - Slow mode and stop mode allow further power reduction
 - High level of modularity
 - Expandable data and program RAM and/or ROM
 - User-definable registers
 - Three parallel execution units: PCU, DAAU, CBU



Embedded DSP Cores (2)

◇ OakDSP core (2)

■ Three parallel execution units

● Program Control Unit (PCU)

- Fetches instructions from the separate program memory.
- Decodes instructions, also performing exception handling and hardware loop control

● Data Address Arithmetic Unit (DAAU)

- Performs all the manipulations required to support the data memory addressing modes.

● Computation/Bit manipulation Unit (CBU)

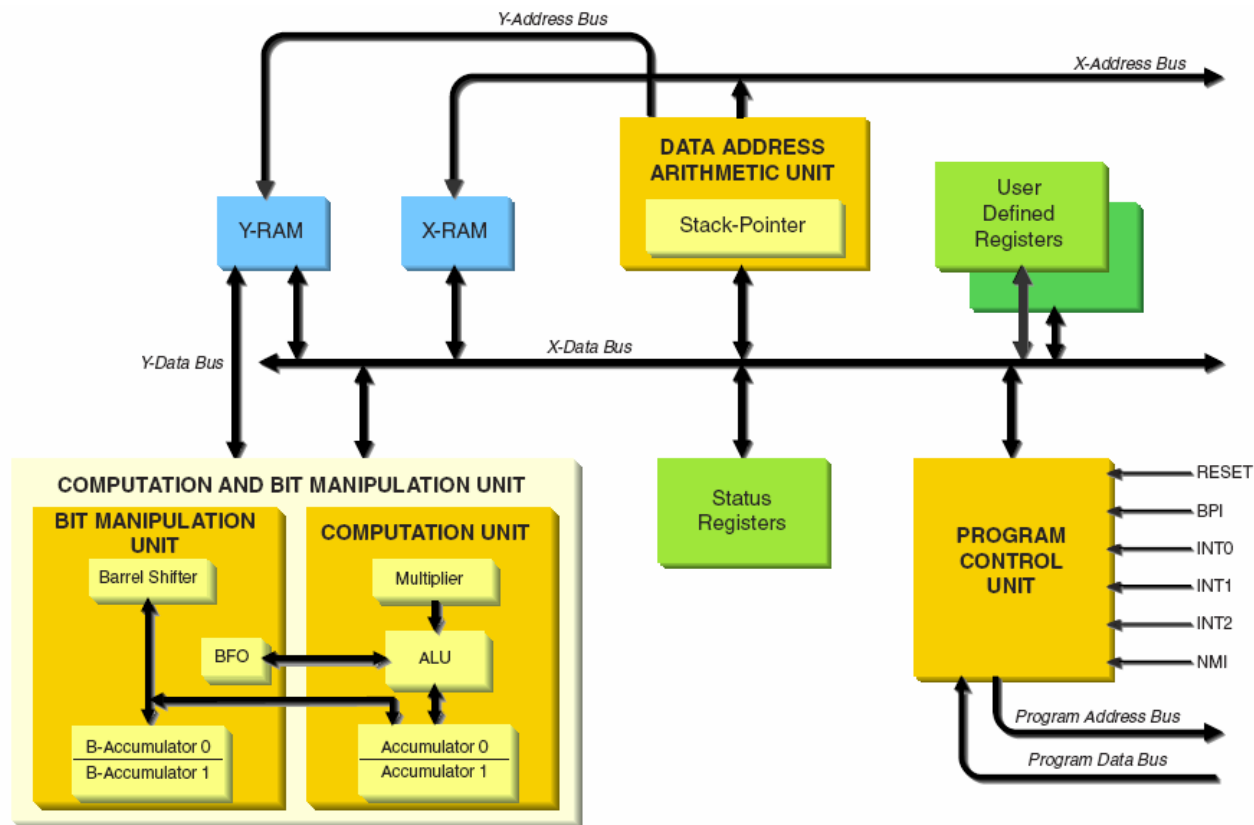
- Contains the Computation Unit (CU) and Bit Manipulation Unit (BMU)
- Single MAC CU consists of a 16x16bit multiplier, a 36bit ALU, and two 36bit accumulators with access to the two additional accumulators of the BMU
- BMU contains a full 36bit barrel shifter, a Bit Field Operation (BFO) unit that includes dedicated hardware for exponent calculation, and two 36bit accumulators



Embedded DSP Cores (3)

◇ OakDSP core (3)

■ Block diagram



Embedded DSP Cores (4)

◇ TeakDSP core (1)

- 16bit fixed-point DSP core based on dual MAC architecture
- Characterized by high performance and high flexibility, as well as rapid data throughput for complex digital signal processing implementations.
- Especially suitable for high-performance & low-power speech and audio processing, multimedia and wireless communications, high-speed modems, advanced telecommunication systems and various embedded control applications



Embedded DSP Cores (5)

◇ TeakDSP core (2)

■ Features

- Performance (operating speed)
 - 180MIPS at 180MHz on 0.25um CMOS 2.5V
- Low power consumption
 - Active mode: using full DSP capabilities
 - Slow mode: linearly reduced clock speeds and current consumption
 - Stop mode: leakage current only
- High level of modularity
 - Expandable data and program RAM and/or ROM & user-definable registers
- Dedicated FFT accelerator
 - 5 cycles for butterfly calculation
- Viterbi built-in accelerator
 - 3 cycles for two add-compare select (ACS)
- Five parallel execution units: PCU, IDU, DAAU, OFU, CBU



Embedded DSP Cores (6)

◇ TeakDSP core (3)

■ Five parallel execution units

● Program Control Unit (PCU)

- Fetches instruction from the separate program memory and passes them to the autonomous Instruction Decode Unit (IDU)

● Data Address Arithmetic Unit (DAAU)

- Performs all the manipulation required to support the various data memory addressing modes
- It has two parallel address lines to each of the three data memory spaces (X, Y and Z for peripherals)

● Operand Fetch Unit (OFU)

- It can read or write, in parallel, two words to or from each of the data memory spaces
- It can transfer four words in parallel to the Computation and Bit manipulation Unit (CBU)
- This minimizes the potential bottleneck due to relatively slow memory transfers.



Embedded DSP Cores (7)

◇ TeakDSP core (4)

■ Five parallel execution units

● Computation/Bit manipulation Unit (CBU)

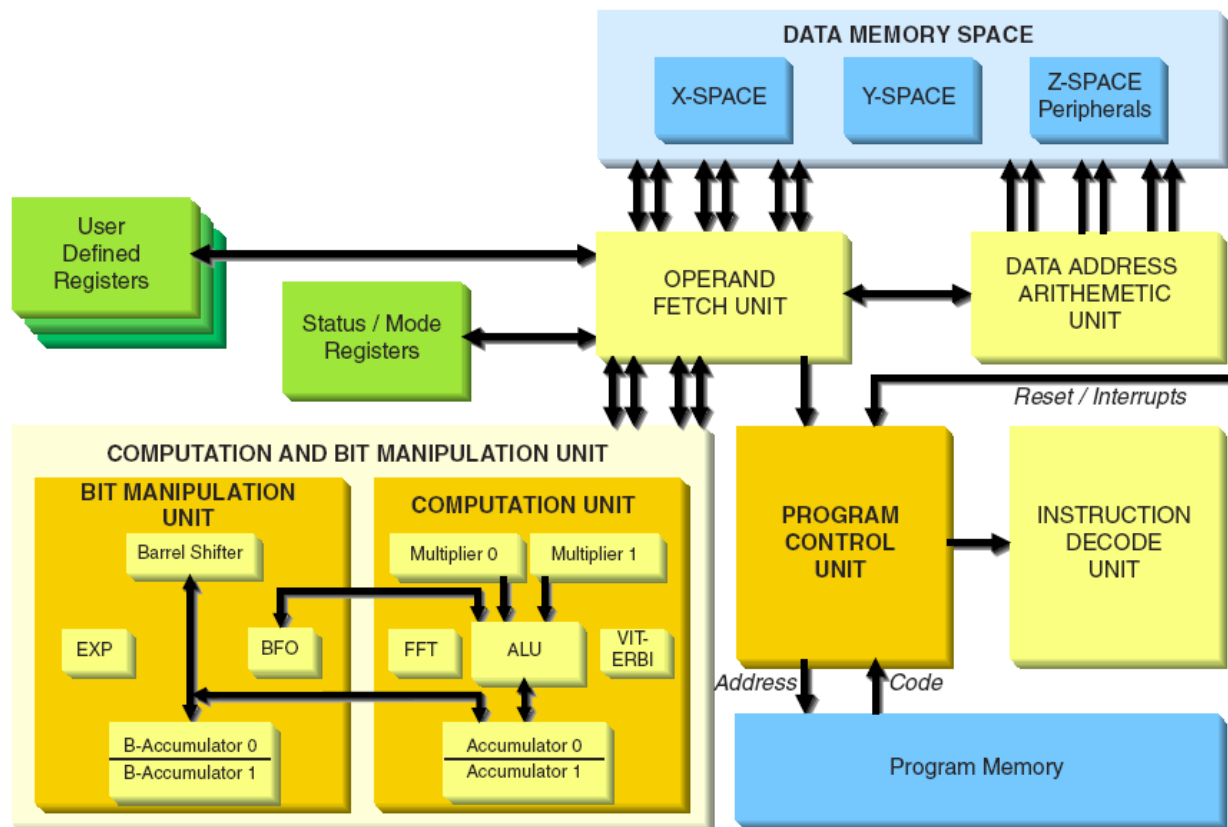
- Contains the Computation Unit (CU) and Bit Manipulation Unit (BMU)
- Two 16x16 bits multipliers
 - Dual-MAC in a single cycle and double precision multiplication
- Four parallel 16bit transfer from data memory
- 40bit ALU and 40bit barrel shifters
- Four independent 40bit accumulators
 - Partitioned into 2x16 bits + 8 guard bits
 - Store/Restore 32bit accumulator part in a single cycle
- Dedicated FFT and Viterbi accelerators
 - 5 cycles for butterfly calculation
 - 3 cycles for two add-compare-select (ACS) operation



Embedded DSP Cores (8)

◇ TeakDSP core (5)

■ Block diagram



Embedded DSP Cores (9)

◇ PalmDSP core family (1)

- 16/20/24bit fixed-point DSP core family based on dual MAC architecture
 - ⊗ Variable data width allows flexibility in the selection of a suitable core member to meet the requirements set by the applications.
- New generation of TeakDSP and OakDSP cores
- Designed for applications that require high throughput and high flexibility.
 - ⊗ Especially suitable for 2.5G and 3G cellular handsets and infrastructure, broadband communication, Voice over IP gateway and consumer multimedia



Embedded DSP Cores (10)

◇ PalmDSP core family (2)

■ Features

- Dual MAC and parallel-functioning architecture
- Execute up to 18 operations in a single cycle
- Supports SIMD and MIMD style instructions
- Low power consumption
 - Active mode: using full DSP capabilities
 - Slow mode: linearly reduced clock speeds and current consumption
 - Stop mode: leakage current only
- High level of modularity
 - Modular word size of ALU, ASU, Barrel-Shifter and Accumulator Reg. file
- Built in FFT support
- Built in Viterbi decoder and trace-back support
- Five parallel execution units: PCU, IDU, DAAU, OFU, CBU



Embedded DSP Cores (11)

◆ PalmDSP core family (3)

■ Block diagram

